



MORE THAN PLATES (PATENT VS MARKETING) · SCRIPT A

The patent describes more than plates

About 3.0 min at ~140 words/min · 414 words · Replace [Your Name]. Speak calmly. One speaker, one focus.

Tip: swap bracketed fields. Cite numbers carefully. End with the cancel / yearly-vote ask.

Mayor Crawford, members of City Council. My name is [Your Name], and I live in Crossville.

Flock sells these cameras as license plate readers. The product, and the company's own patent, go much further than plates.

U.S. Patent US11416545B1 is assigned to Flock Group Inc. The inventors include CEO Garrett Langley. It was granted August 16, 2022. The title is not "license plate reader." It is a system for searching video across a network of cameras by what is in the footage.

Read what Flock told the Patent Office. The system uses AI to sort what the cameras see. Vehicles are one category. People are another. For people, the filing describes classifying "male, female, race, etc.," clothing types, and estimated height and weight. In one scenario in the patent, investigators get "face recognition data points" from one camera and search other cameras for matching face recognition data points. That means matching people across cameras, not reading a plate.

Flock's public pages tell cities plate-reader products "do not use facial recognition" and "do not identify race, gender, or ethnicity." The company still advertises search for people by clothing, like "man in blue shirt," live video cameras that can lock onto a person and follow them, and car search by make, model, color, and markings when a plate is missing.

Here is what happened in Oshkosh, Wisconsin. In April 2026, that city council asked Flock in open meeting whether the system creates a heat map of a vehicle's movement. Flock's chief information security officer said no, that is not available. The next morning, Police Chief Dean Smith said the system does create heat maps. Council killed the contract that same day, seven to zero, and took the cameras down.

That patent does not prove Crossville's cameras are doing all of this today. It does prove what Flock told the government it can build. So do not buy the "just plates" pitch, and do not trust answers you cannot check. Ask in writing which people-search features are on or off, then cancel until Crossville gets answers residents can verify.

THE ASK (say this)

My ask is clear. Cancel Crossville's Flock contract and remove the cameras. Require a public hearing and a recorded City Council vote before any surveillance equipment is ever proposed again. If you will not cancel yet, take a recorded yes-or-no vote each year before any renewal, with no automatic renewal that skips a public vote, and publish monthly use reports the public can check. Thank you. I yield.

Speaker prep visual (do not read aloud)

US 11,416,545 B1

7

Also, a nearby traffic light intersection outside the geographic footprint of the neighborhood 10 includes a traffic monitoring camera TMC-1 that captures video of any and all vehicles that pass through the intersection, including a picture of the associated license plates (the traffic monitoring camera forms its own component network within the dynamic surveillance network). And, nearby convenience stores outside the geographic footprint of the neighborhood 10 use commercial security systems CSS-1, CSS-2 with video surveillance devices (each commercial security system forming its own component network within the dynamic surveillance network).

A pedestrian with a portable recording device PRD-1 frequently walks through neighborhood 10. All of the exemplary surveillance devices CSS-1, CSS-2, HSS-1, HSS-2, NSD-1 through NSD-8, PRD-1 and TMC-1 are configured to record video and, optionally, upload the recorded footage to an embodiment of a dynamic surveillance and object-based query ("DSOQ") system according to the solution. Further, depending on a particular embodiment, exemplary surveillance devices may upload recorded footage for a first time based on an "opt-in" permission and, subsequently, upload recorded footage at a request or command of the DSOQ embodiment. In this way, embodiments of the solution may avoid upload and management of irrelevant or unneeded video footage at the DSOQ.

FIG. 1A represents activity in the neighborhood 10 at time window "n." During time window "n," a red car #1 entered neighborhood 10 between Block 5 and Block 8, drove straight through, and exited neighborhood 10 between Block 4 and Block 7. During the same time window "n," a blue car #1 entered the geographic footprint of neighborhood 10 between Block 1 and Block 4, turned left between Block 4 and Block 5, drove straight through, and exited neighborhood 10 between Block 7 and Block 8.

Red car #1 was captured as video content by TMC-1, NSD-4, PRD-1, and NSD-3. Similarly, blue car #1 was captured as video content by NSD-1, NSD-5 and NSD-7. Notably, the pedestrian manning PRD-1 didn't video blue car #1 due to recognizing the car and its driver as a local resident. By contrast, the pedestrian manning PRD-1 recorded video of red car #1, whether unintentionally while recording something else, or intentionally when noticing that the car was being driven slowly by a person the pedestrian didn't recognize.

FIG. 1B represents activity in neighborhood 10 at time window "n+1." During time window "n+1," a blue car #2 entered neighborhood 10 between Block 4 and Block 5, drove straight through, and exited neighborhood 10 between Block 7 and Block 8. Blue car #2 was captured as video content by surveillance devices CSS-1 (blue car #2 stopped for gas), NSD-5 and NSD-7.

FIG. 1C represents activity in neighborhood 10 at time window "n+2." During time window "n+2," red car #2 entered the geographic footprint of neighborhood 10 between Block 1 and Block 4, turned left between Block 4 and Block 5, drove straight through, and exited neighborhood 10 between Block 7 and Block 8. Red car #2 was captured as video content by NSD-1, NSD-5 and NSD-7.

Also during time window "n+2," red car #1 makes a second appearance in neighborhood 10. Red car #1 enters neighborhood 10 between Block 4 and Block 5, turned right between Block 5 and Block 8, turned right again between Block 5 and Block 6, then exited the geographic footprint of neighborhood 10 by turning left between Block 5 and Block 6. Red car #1 was captured as video content during time

8

window "n+2" by CSS-1 (driver of red car #1 filled up red car #1 with gas and bought a candy bar). NSD-1, NSD-4 and TMC-1.

Referring to each and all of FIGS. 1A-1C, which illustrate different time windows and a dynamic geographic footprint relevant to a burglar investigation at house X, video content relevant to the burglar investigation may have been captured by any one or more of the various surveillance devices of the various component networks. As previously noted, investigators relying on the prior art would have to independently seek out and watch footage from each surveillance device of each component network over a given window of time—no piece of footage could go unwatched lest a relevant content be missed. In this way, users of the prior art may spend significant amounts of time identifying video sources and watching footage in an effort to identify relevant content. Advantageously, and as will become more apparent from the illustrations and description that follows, embodiments of a DSOQ system may aggregate video footage collected from multiple independent component network sources, analyze the footage, and store the footage based on the results of the analysis so that the footage may be queried in view of content and not just time/location.

Returning to the application scenario illustrated by FIGS. 1A-1C, investigators of the burglary event may put out a public notice seeking any information that may be relevant to the burglary. In response, the pedestrian associated with PRD-1 in FIG. 1A may let investigators know that she had recently witnessed an out of place car in neighborhood 10. Subsequently, an upload to the DSOQ system of the video taken on PRD-1 during time window "n" of red car #1 would get analyzed such that the red car #1 is identified as to make and color.

Focusing on the red car #1 "lead," the investigators may leverage the DSOQ system to return any footage captured by any of the NSD cameras of the neighborhood watch system that contains red car #1, dating back to the time stamp of the video captured by PRD-1. The DSOQ system may then return footage captured by NSD-4 and NSD-3 during time window "n" as well as footage captured by NSD-1 and NSD-4 during time window "n+2." Notably, and advantageously, the investigators could rule out any footage captured during time window "n+1" as no red car was identified by the DSOQ system in any content associated with video footage taken during that window. Further, depending on the statistical confidence of the object detection for footage taken during time window "n+2," video footage containing red car #2 may also be ruled out for viewing.

Using information from viewing of all footage with red car #1, the investigators may be able to narrow the time window during which red car #1 was in or around neighborhood 10. Dynamically increasing the zone of interest to other component networks within the dynamic surveillance network and expanding the query for video footage with content having a red car, the investigators may leverage the DSOQ system to identify video footage taken by CSS-2 when the driver of red car #1 purchased gas and a candy bar.

Advantageously, because the footage from CSS-2 may be queried and returned by the DSOQ system based on having the red car #1, investigators may avoid watching untold hours of irrelevant footage captured by CSS-2. With the relevant footage returned from CSS-2, the investigators may be able to identify the face of the driver of red car #1 when he purchased the candy bar. Now, with face recognition data points, the investigators may be able to query the DSOQ system for any footage captured by HSS-1 or HSS-2 having a person with matching face recognition data points. In this

US11416545B1 specification: "face recognition data points" cross-camera query example

Speaker prep visual (do not read aloud)

U.S. Patent

Aug. 16, 2022

Sheet 8 of 13

US 11,416,545 B1

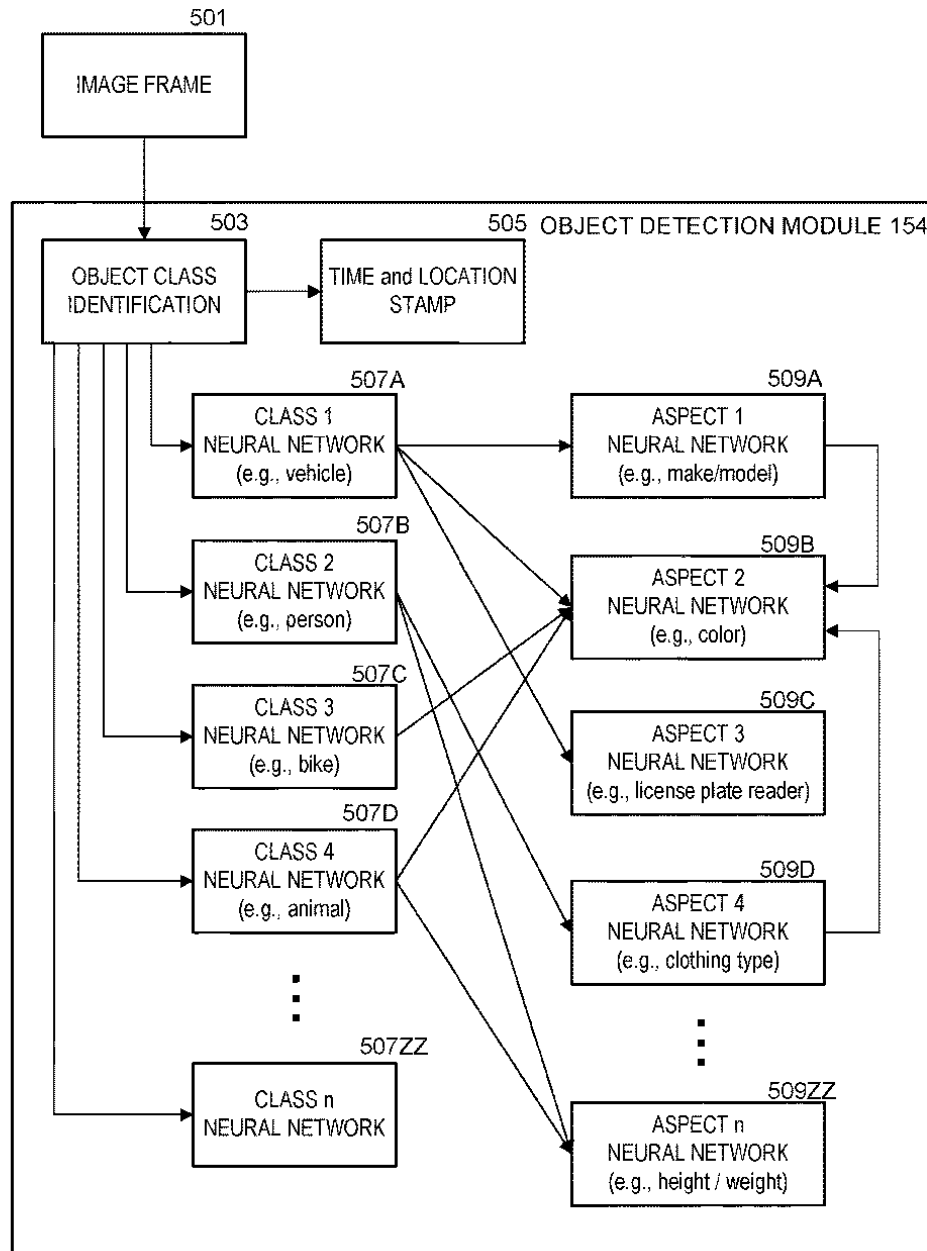


FIG. 5A

US11416545B1 FIG. 5A: person class + clothing / height-weight networks (patent drawing)

Speaker prep visual (do not read aloud)

US 11,416,545 B1

19

in association with the video footage is data indicative of content identified by the object detection module 154.

The object detection module 154 begins by identifying objects of interest in a given image frame 501. Objects of interest may be identified based on feature learning techniques. Methodologies for identifying relevant features of objects may be, but are not limited to, Shi-Tomasi corner detection algorithms or Harris Corner Detection Method, as would be understood by one of ordinary skill in the art. With an object identified in the image frame 501, the object detection module 154 leverages a cascade of neural networks 507, 509 to identify aspects of the object.

For example, if the object class of the identified object is that of a human being, then the object detection module 154 may further analyze the image 501 using a neural network module 507B configured to identify different classes of people (male, female, race, etc.). Subsequently, with aspects of the person identified, the object detection module 154 may leverage other, more detailed neural networks 507, 509 to further identify aspects of the person such as, but not limited to, a neural network 509D for identifying clothing types (jacket, pants, shorts, hat, etc.) and a neural network 509ZZ for gauging height and weight.

As other examples, a neural network 507A may be assigned to determine if an image 501 contains a vehicle class of objects. Another neural network 507C may be assigned to determine if an image 501 contains a bicycle class of objects. Similarly, another neural network 507D may be assigned to determine if an image 501 contains an animal class of objects. As understood by one of ordinary skill in the art, other classes of objects are possible and are included within the scope of this disclosure.

For these class objects, other neural networks 509A, 509B, 509C, 509D, and 509ZZ may be assigned to determine if each class of object found by the class neural networks 507 have aspects, such as, but not limited to, make/model of vehicles; colors; license plate readers/recognition modules; clothing types; and/or height/weight of people objects, just to name a few. As understood by one of ordinary skill in the art, other aspects for each class of objects are possible and are included within the scope of this disclosure.

With the outputs from the relevant neural networks 507/509, the object detection module 154 may work to store the outputs in database 120 in association with the image frame 501 (and, by extension, the video footage segment that contains the image frame 501) along with the time and location data. In this way, the database may be configured to be leveraged by the historical analysis module 162 and/or search module 156 and/or alert triggering module 158 as previously described.

Referring now to FIG. 5B, this figure is a logical flow chart for a method 507', 509' for training a neural network 507, 509 to detect objects in images 501. This method is usually performed by the DSOQ server 103. Once the method 507', 509' is complete, the server 103 may send the calculated weights for the one or more neural networks [acting as filter(s)] to the object detection module 154 of FIG. 5A.

One goal of the machine-based object detection method 507', 509' as illustrated in FIG. 5B is to be able to input an image (Step 510) and output a label of the objects in that image (e.g. car, dog, etc.) (Step 540). Here, the method 507', 509' may train each neural network 507, 509 of FIG. 5A its particular class or aspect as mentioned above in connection with FIG. 5A. A common technique for creating an object detection system is to train one or more convolutional neural

20

networks via machine learning, where the model's accuracy is improved by being provided examples of correct results.

To accomplish this, a dataset must first be compiled with example images where the objects in the image are labeled prior to training (Step 510). All objects that are desired to be detected must be known and incorporated into the model and example data beforehand. Many existing datasets are available for use, such as COCO and Open Images; otherwise, one may label his/her own images. For training, it is recommended to have at least 50,000 images/video segments of each object to be detected.

With this dataset in hand, a convolutional neural network may be trained to learn how to correctly detect objects in an image. As understood by one of ordinary skill in the art, a neural network usually includes a graph of matrix operations where some matrices have adjustable variables (i.e. weights). One goal of the training process is to be able to adjust these weights so that certain input values (i.e. images) result in certain output values (i.e. correct labels). Existing network architectures exist, such as Faster RCNN and YOLO, and are at least two preferred architectures as of this writing.

The training process for a neural network may include inputting labeled images/video segments into a neural network (Step 510), evaluating the image through the neural network (Step 515), and comparing the predicted label generated by the neural network with the actual label (=label made prior to training) (Decision Step 520).

If there is a difference between the expected and actual values (i.e. loss) as calculated in decision Step 520 where the prediction is wrong/off the method goes to step 525. If there is no difference between the expected and actual value (i.e. no loss), then the method proceeds to decision Step 530.

In Step 525, the difference between the expected and actual values is calculated and applied to the matrix weights of the neural network such that the loss is reduced for the next iteration of Step 520, improving the accuracy of the neural network.

After Step 525, in decision step 530, it is determined if a desired total accuracy of the neural network has been achieved. If the inquiry to decision Step 530 is positive meaning that the desired accuracy was achieved in Step 520, then the "Yes" branch is followed to Step 535.

If the inquiry to decision Step 530 is negative—meaning that the desired total accuracy based on the comparison in Step 520 has not been achieved, then the "No" branch is followed back to Step 510 wherein the next training image in the training dataset is inputted into the neural network. In Step 535, the network matrix weights are saved. In step 540, the saved network matrix weights may be used to predict labels for objects 146 (i.e. license plates, cars, people, etc.) in new images (not from the training dataset).

This process is then repeated (returned to Step 510) across the entire dataset until the weights begin to converge. A neural network 507, 509 with converged weights provides consistent accuracy across the entire dataset, yielding an object detection model in which you can input a new image and detect the objects in that image with an expected accuracy.

Once the neural network 507, 509 achieves a desired, selected accuracy, the DSOQ server 103 may transmit the saved network matrix weights to the object detection module 154 so that the one or more neural networks 507, 509 that form the object detection module 154 may be updated.

For a specific neural network training example, see a journal article entitled, "Faster RCNN: Towards Real-Time Object Detection with Region Proposal Networks," authored

US11416545B1 specification: "male, female, race, etc." people classification language